

Artificial Intelligence & RAG Technology Report

1. Executive Summary

Artificial Intelligence (AI) has revolutionized how organizations handle unstructured text documents. Retrieval-Augmented Generation (RAG) combines search retrieval algorithms with generative Large Language Models (LLMs) to deliver grounded, accurate, and verifiable answers from proprietary PDF documents and knowledge bases.

2. How Question Answering Works

The PDF Question Answering engine operates through three main stages:

1. Document Ingestion & Chunking: PDF files are parsed into semantic text passages while tracking page numbers.
2. Vector Indexing: Text chunks are converted into numerical representations using TF-IDF or vector embeddings.
3. Context Retrieval & Answer Generation: When a user asks a question, top matching context chunks are retrieved and passed to the AI model to generate a precise answer with source citations.

3. Key System Features

- Page-Level Citations: Users can verify exact page numbers for every extracted fact.
- Dual Execution Modes: Supports offline extractive answering as well as cloud-based Gemini LLM generation.
- Interactive UI: Built with Streamlit for clean document metrics and interactive conversation history.

4. Recommendations & Future Outlook

Organizations adopting AI PDF QA systems should ensure proper document OCR pre-processing for scanned PDFs, implement chunk size optimization, and enforce security policies around document privacy.